FILTER DESIGN TEST CASE

SAW (Surface Acoustic Wave) - Band 1 (2100 MHz) for Automotive V2X Communication
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a SAW (Surface Acoustic Wave) filter targeting Band 1 (2100 MHz) for Automotive V2X Communication. The filter is designed on AlN on Si substrate with a target center frequency of 4290.9 MHz and bandwidth of 265.2 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Automotive V2X Communication module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: SAW (Surface Acoustic Wave)
- Frequency Band: Band 1 (2100 MHz)
- Application: Automotive V2X Communication
- Substrate: AlN on Si
- Package: SiP (System in Package)
- Design Tool: CST Studio Suite 2024
- Design Center: Helsinki 5G Lab

This specification applies to all variants of the SAW (Surface Acoustic Wave) design targeting Band 1 (2100 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 4290.9 MHz
Bandwidth (3 dB): 265.2 MHz
Insertion Loss Target: <= 3.2 dB
Return Loss Target: >= 22.0 dB
Out-of-Band Rejection: >= 35.0 dB
Isolation (Duplexer): >= 37.0 dB
Q Factor: 955
Temperature Coefficient: -12.8 ppm/°C
Die Size: 1.3 x 0.8 mm
Wafer Size: 8-inch
Number of Resonators: 11
Electrode Material: W

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the SAW (Surface Acoustic Wave) template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 4290.9 MHz, BW 265.2 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, AlN on Si).
10. Perform on-wafer probing using Rohde & Schwarz ZNB40.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 12873 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.2 dB
- Return loss in passband shall be >= 22.0 dB
- Out-of-band rejection at +/- 398 MHz offset >= 35.0 dB
- Group delay variation across passband shall be <= 10 ns
- Temperature drift over -40°C to +85°C shall be <= 6.9 MHz
- Power handling shall be >= +29 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 93%

6. TEST RESULTS

Measurement Date: 2024-03-25
Devices Tested: 29
Insertion Loss (meas): 2.89 dB
Return Loss (meas): 23.9 dB
Rejection (meas): 38.1 dB
Isolation (meas): 42.1 dB
Q Factor (meas): 955
Group Delay Variation: 7.2 ns
Power Handling: +35 dBm
Die Yield: 85.5%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

a) The SAW (Surface Acoustic Wave) design demonstrated excellent performance for Band 1 (2100 MHz).
b) Insertion loss is marginally above target; electrode thickness optimization recommended.
c) Out-of-band rejection exceeded the minimum requirement.
d) Group delay flatness improved compared to previous revision.
e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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Approver:	Dr. James Liu	Date: 2024-04-03

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